CHITRANG POTDAR

Backend & AI Engineer

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PROFESSIONAL SUMMARY

Engineer who starts from the core problem and picks the simplest architecture that holds up, so systems stay maintainable long after release. Builds backends that scale and stay secure under real load, with correctness, concurrency, and failure modes designed in from the first schema. Ships agentic AI systems as structured, observable, self-correcting workflows rather than one-shot prompts, owning the full lifecycle from design to deployment.

TECHNICAL SKILLS

Languages: Python, Rust, JavaScript, SQL
Backend / Systems: FastAPI, Flask, REST APIs, Celery, RESP2 Protocol, Rate Limiting, Hybrid Encryption (AES/RSA)
Databases & Caching: PostgreSQL, Redis
AI & Agent Frameworks: Claude API, LlamaIndex, LangChain, LangGraph, MCP
Cloud & DevOps: AWS (Lambda, API Gateway, S3, EC2), Docker, GitHub Actions, CI/CD
Integrations: SendGrid, Vimeo, Web Push (VAPID), Google OAuth, Facebook OAuth, Firebase Admin
Observability & Testing: Grafana, Loki, Promtail, LangSmith, Pytest, Playwright

PROFESSIONAL EXPERIENCE

Backend Engineer · ArrayPointer Pvt. Ltd., Pune **Oct 2025 – Present**

Investment Operations Platform

Flask · PostgreSQL · Redis · Celery · Docker · GitHub Actions · AWS EC2 · Grafana · Loki · Promtail · Pytest · Playwright

- Architected a multi-tenant backend with query-level data isolation across users, portfolios, and organizations on a shared PostgreSQL schema; added Redis session caching to cut redundant DB round-trips, sustaining 80%+ Pytest coverage.
- Built REST APIs for portfolio review, rejected-order tracking, and financial calculators; integrated the Jira API to auto-create issues from application events, eliminating manual reporting.
- Implemented granular RBAC mapping features to domains, with role-level CRUD access per feature.
- Automated NSE rejected-order processing via Python API integrations and Playwright-driven end-to-end client workflows, saving ~2–3 hrs/week.
- Established observability (Promtail → Loki → Grafana) for structured visibility into service health and error trends.
- Built a domain-specific MCP server exposing platform data to LLM clients, annotating tools with domain terminology so models resolve business concepts without raw schema knowledge.
- Implemented async email delivery (Redis + Celery) and VAPID Web Push for real-time single-user alerts, with in-UI failure notifications for undelivered email.
- Extended task management with hierarchical ownership and access control across organizational levels; delivered task reporting, employee performance analytics, CRM workflows, and session tracking, with task/CRM API restrictions enforcing workflow and availability rules.

LMS & CMS Platform — Jaro Education

FastAPI · PostgreSQL · SendGrid · Vimeo · AWS S3 · Modular Monolith

- Designed a modular-monolith LMS/CMS backend on FastAPI and PostgreSQL, supporting 10K+ students at ~300 RPS with SendGrid email-verified authentication.
- Built admin modules for course creation and student management, integrating Vimeo for video upload and streaming.
- Developed student-facing APIs for course access and certificate retrieval, with AWS S3 backing certificate storage.

MahaNRI — NRI Social & Community Platform (Backend Lead)

FastAPI · PostgreSQL · Redis · Firebase

- Led backend development of a Maharashtra Government social platform for the NRI community, serving multiple applications from one shared backend; directed developers/interns through task breakdown, technical guidance, code reviews, PR management, and merge decisions.
- Designed REST APIs across application modules, coordinating API contracts with frontend teams and translating client requirements into backend tasks.
- Implemented backend-side hybrid encryption (AES/RSA) for end-to-end API security: unwrapped the client's session-scoped symmetric key with the server's RSA private key, then used it to decrypt request payloads and encrypt responses.
- Implemented Redis-based rate limiting across the shared multi-app backend services.
- Built Firebase OTP verification with backend token validation via firebase-admin, plus Google and Facebook OAuth sign-in.
- Built FCM-based push notifications delivering real-time alerts to web and Android clients.
- Built LinkedIn-style search returning grouped results across People, Posts, Organizations, and Events.
- Owned production deployments and database migrations across all application modules.

PROJECTS

SOLID-Agent-Harness — Multi-Agent Code Analysis Tool

Node.js · LangChain · LangGraph | github.com/Dev-Chitrang/SOLID-Agent-Harness

- Built a language-agnostic CLI platform running structured, graph-based LLM workflows for repository analysis, documentation generation, and code quality: six commands (audit, bugs, docs, readme, review, explain) plus five composite workflows.
- Implemented a critic-loop pattern per graph (Agent → Critic → conditional retry) that exits on a quality-score threshold or iteration cap, improving reliability over single-pass agent calls.
- Designed composite commands using LangGraph parallel fan-out with a barrier-join summary node, running multiple analysis graphs concurrently and synthesizing one executive report.
- Built a provider-abstraction layer (OpenAI, Anthropic, Gemini, NVIDIA) swappable without changing graph code, with an interactive TUI for provider/command/model selection.
- Architected registry-driven command dispatch so new commands ship via a folder and registry entry alone, with zero changes to core harness or TUI files.

Redis Clone in Rust — Async In-Memory Key-Value Server

Rust · Tokio · RESP2 Protocol · Docker · Python (E2E Testing) | github.com/Dev-Chitrang/Redis-In-Rust

- Built an async TCP in-memory key-value server implementing RESP2 (pipelined and fragmented frames) over a 64-shard RwLock keyspace with deadlock-free ordered multi-shard locking, sustaining up to ~11.8K write RPS multi-threaded.
- Implemented typed storage (String, Hash, List, Set, and a size-augmented AVL-tree SortedSet for O(log n) rank lookups), lazy plus background TTL expiration, and channel/glob-pattern Pub/Sub.
- Engineered AOF durability with startup replay and tail-repair, atomic BGREWRITEAOF compaction, and a batched group-commit writer (1 ms / 64-op) that raised synchronous always-fsync write throughput 9.37× (up to 1,774 RPS at 50 clients) and cut P99 latency 85.1% (336 ms → 50.1 ms).
- Built a Python end-to-end suite (redis-py and raw sockets) validating concurrency and atomicity, plus a 1,080-run redis-benchmark harness sweeping execution modes, client counts, pipeline depths, and AOF policies.

EDUCATION

B.E. Computer Engineering — 8.65 CGPA · Savitribai Phule Pune University, Pune 2021 – 2025

RESEARCH & PUBLICATIONS

Corporate Credit Rating Analysis using AI & Sentiment Analysis — IJIRCCE (Impact Factor 8.65) Nov 2024